Dynamic Taxonomy a Bridge from DeFi to TradFi

Iqbal Mabruri[1], Carles Cerqueda[1], Christopher Jack[1], Alexis Lui[1], Yue Wu[1], Wenbin Wu[1], Vincenzo Di Perna[1,2], Keith Bear[1], and Bryan Zhang[1]

[1]Cambridge Centre for Alternative Finance, Cambridge Judge Business School, Cambridge University
[2]University of Urbino - Italy

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Abstract

Blockchain technology has seen remarkable growth over the past decade, with Ethereum emerging as a leading platform due to its smart contract capabilities, which support various blockchain-based applications. Among these applications, decentralised finance (DeFi) has enabled both decentralised and semi-decentralised versions of traditional financial services, including P2P lending, index funds, and exchanges. By 2024, the DeFi ecosystem's value has reached approximately $167.52 billion, with continued expansion expected. As DeFi services increasingly grow in popularity, they start to offer more services similar to traditional finance (TradFi), a structured taxonomy is crucial to improve regulatory clarity, support compliance, and enhance integration with existing financial infrastructures. This paper examines both DeFi and TradFi taxonomies to propose a dynamic framework that bridges these financial systems, offering distinct advantages for regulators and DeFi practitioners. Our analysis reveals that DeFi encompasses both unique innovations and services resembling those in TradFi, and we identify clear connections across DeFi categories that can inform a regulatory framework focused on key elements: activities, actors, assets, and the underlying architecture of DeFi protocols.

Keywords— Decentralised Finance (DeFi), Blockchain, Financial Infrastructure, Taxonomy

Introduction

Blockchain technology continues to reshape the global economic landscape, beginning with Bitcoin's introduction as the first censorship-resistant, public, and permissionless blockchain. Bitcoin established blockchain's use in payments, while Ethereum further expanded its potential by enabling smart contracts—self-executing code that facilitates a range of financial applications. This innovation paved the way for Decentralised Finance (DeFi), which leverages blockchain technology by issuing its own tokens and operating via protocols based on smart contracts. Today, a plethora of DeFi protocols make up the decentralised finance ecosystem, aiming to provide financial services similar to those of traditional finance (TradFi) — such as banks, financial institutions, and regulated markets — while leveraging the technological novelty of blockchain to offer a more sophisticated service than their TradFi counterparts.

Early DeFi projects often addressed specific needs within individual blockchains, due to technological constraints, differing consensus mechanisms, and the nascent interoperability standards within the blockchain ecosystem. Stablecoin - digital assets designed to maintain stable value, typically against fiat currencies (for Alternative Finance, 2024) - and Decentralised Exchanges (DEXs) are among the early DeFi services, which offer digital asset liquidity on blockchains. These innovations were created to tackle price volatility and liquidity, which still are a topic to be

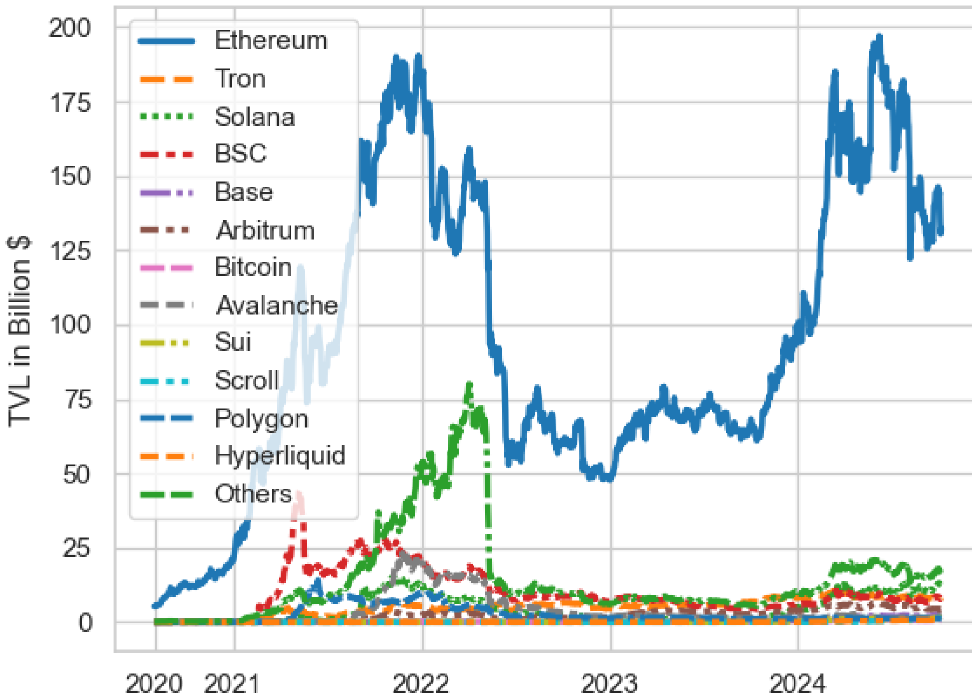

Figure 1: Top 10 Blockchains TVL 2020-2024

solved in the digital asset's market. With Ethereum's transition to the Proof of Stake (PoS) consensus mechanism (Foundation, 2020), new challenges and opportunities arose for DeFi. One key challenge was reduced liquidity, as PoS consensus mechanism require locking up significant amounts of tokens, effectively removing them from circulation. Furthermore, stakers gained greater control over which transactions to prioritise processing based on varying incentives, leading to higher transaction fees - a phenomenon known as Maximum Extractable Value (MEV) (Daian et al., 2019). In response, DeFi introduced solutions like liquid staking, allowing users to stake smaller amounts without the need to run a validator node, thereby participating in staking while preserving liquidity for other uses such as lending and borrowing. By 2024, the Total Value Locked (TVL) across over 200 DeFi protocols has reached approximately $167.52 billion, underscoring DeFi's rapid growth and its expanding role in financial innovation (DefiLlama, 2020b).

Although many DeFi services were initially developed to address the limitations of blockchain platforms, they have subsequently demonstrated the potential to provide a novel form of financial services that leverage blockchain technological capabilities. This evolution has resulted in DeFi services progressively offer services similar to TradFi, the need for a comprehensive taxonomy becomes critical for regulators and policymakers to identify risks and manage vulnerabilities within the system. However, the rapid evolution of DeFi has made taxonomy development challenging, leading many authors to focus narrowly on its technological aspects and layered application models (Schär, 2021; Jensen et al., 2021; Brühl, 2021).

Despite numerous attempts to classify DeFi a unified taxonomy encompassing the full ecosystem remains elusive (Chen and Bellavitis, 2020), (Moncada et al., 2021), (Nelaturu et al., 2022), (Xu and Xu, 2022), (Gogol et al., 2024). Moreover, many scholars treat DeFi as a novel financial service distinct from TradFi, even though many DeFi protocols closely resemble their TradFi counterparts. This difference stems from DeFi's reliance on blockchain-based distributed ledgers versus TradFi's centralised infrastructure. The lack of a universal taxonomy hinders effective communication among stakeholders, including developers, users, and regulators, impeding regulatory progress (Yousuf et al., 2024; Mohan, 2022). Furthermore, the rapid development of new DeFi protocols and products often outpaces existing taxonomic frameworks, rendering them obsolete and lacking the most recent innovations or modifications in DeFi applications (Aquilina et al., 2024; Yousuf et al., 2024). Therefore, there is a need for a comprehensive taxonomy that categorises DeFi protocols based on their functional, technological, and unique attributes.

In this paper, we present a dynamic DeFi taxonomy that seeks to bridge the divide between DeFi and TradFi frameworks. We address three core questions:

1. How can a taxonomy effectively support DeFi's dynamic nature while remaining adaptable to continuous developments?
2. What approaches best facilitate DeFi and TradFi interoperability, fostering seamless interaction between these two financial systems?
3. How can this taxonomy serve as a regulatory tool to support transparency, security, and compliance without stifling innovation in DeFi?

The key findings from this study are as follows:

- While the application of smart contracts to provide financial services is central to DeFi's innovation, the ecosystem encompasses both unique services and those closely resembling traditional finance (TradFi) offerings.
- Unique services within DeFi include liquid staking, yield farming, asset tokenisation—covering both real-world and digital assets—and cross-chain asset bridging. In contrast, services like peer-to-peer lending, collateralised loans, index funds, and options closely mirror their TradFi counterparts.

These insights highlight the need for a regulatory framework. By aligning DeFi categories with key regulatory considerations—specifically activities, participants, assets, and the underlying architecture of DeFi protocols—such a framework could be effectively developed.

The rest of the paper is structured as following. Section 2 will present the literature studies on DeFi taxonomy. Section 3 describes the approach to construct our taxonomy. Section 4 describe context about DeFi ecosystem, categories and use sage of this taxonomy. In Section 5, we give direction of future works following this taxonomy. Finally Section 6, conclude this paper.

Related Works

The absence of a unified taxonomy clearly connecting DeFi and TradFi poses challenges for regulators complicating efforts to classify and oversee DeFi projects within existing financial frameworks. This gap in regulatory clarity leaves some DeFi projects unregulated or inadequately regulated (Yousuf et al., 2024), (Mohan, 2022).

Schär was the first to decompose the taxonomy into multiple application layers (Schär, 2021), (Jensen et al., 2021), (Brühl, 2021), focusing primarily on technological distinctions and proposing a layered stack model as shown in Figure 2: the blockchain platform as the base layer for handling settlements; Layer 1, the Asset layer, which represents the token standards used; Layer 2, the Protocol layer, where DeFi is categorised based on activities such as exchanges, lending, and derivatives; the third layer, the Application layer, which provides the interface for users; and finally, the Aggregator layer, which consolidates multiple applications into a single platform (Schär, 2021).

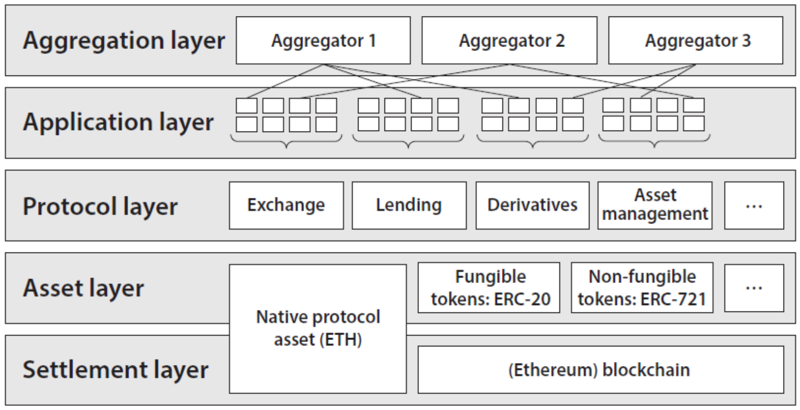

Figure 2: Schär's Model(Schär, 2021)

Following the publication of Schär's model, it gained widespread acceptance and was expanded by subsequent studies to include additional activities, such as Algorithmic Market Makers and Automated Asset Management. (Jensen et al., 2021). Other frameworks simplify DeFi into two main structures: Distributed Ledger Technology (DLT) infrastructure and Decentralised Applications (dApps) (Brühl, 2021).

Several other authors (Chen and Bellavitis, 2020), (Moncada et al., 2021),(Nelaturu et al., 2022), (Xu and Xu, 2022), and (Gogol et al., 2024) have taken a qualitative approach, analysing DeFi's business models by comparing protocol white paper promises to their actual smart contract implementations. This method identifies core components within DeFi protocols, yet no standardised business model has been universally accepted. As illustrated in Table 1, no author to date has reached the same conclusion regarding what DeFi protocol categories exist apart from identifying that DeFi must run on top of blockchain infrastructure.

Table 1: Summary Past Qualitative Approach DeFi Taxonomy Papers.

Title	**Summary**	**Identified DeFi Type Classification**
Blockchain disruption and decentralized finance: The rise of decentralized business models (Chen and Bellavitis, 2020)	The paper examines the inherent properties of blockchain technology, such as decentralisation, transparency, immutability, and distributed trust, to identify the resulting business model.	Decentralised currencies, Decentralised Payment, Decentralised fundraising, Decentralised contracting
Next generation blockchain-based financial services (Moncada et al., 2021)	The paper explores key areas of blockchain, including scalability solutions, interoperability, security enhancements, decentralisation, and use case applications.	Lending & Borrowing, Exchange Trading, Asset Management, Derivatives, and Stablecoin

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Table 1: Summary Past Qualitative Approach DeFi Taxonomy Papers. (Continued)

A Review of Blockchain in Fintech: Taxonomy, Challenges, and Future Directions(Nelaturu et al., 2022)	This paper reviews blockchain in general and provides a taxonomy for DeFi in comparison with fintech, covering its applications across various verticals such as digital payments, investments, lending, insurance, crowdfunding, and loyalty programmes.	Digital Identity, Payments, Digital Currencies, Decentralised Exchanges, Protocols for Loanable Funds, Derivatives, Decentralised Oracles, Decentralised Storage, Services, Corporate Governance, Crowdfunding
A Short Survey on Business Models of Decentralized Finance (DeFi) Protocols (Xu and Xu, 2022)	This paper examines different approaches used by DeFi protocols to generate revenue, sustain operations, and create value for their users. It includes discussions on how these protocols maintain financial stability, trust, and economic soundness within a decentralised framework.	Protocols for loanable Funds, Decentralised Exchanges, Yield Aggregators
SoK: Decentralized Finance (DeFi) - Fundamentals, Taxonomy and Risks (Gogol et al., 2024)	The taxonomy is presented in two dimensions—algorithm and network architecture—separating token classification from protocol classification. This structure enables a clearer understanding of how different protocols operate and the risks associated with them.	Trading, Lending, Asset Management, Synthetic (Pegged) Tokens, Liquidity Pools, Aggregator, Blockchain Interoperability, Liquidity Pools

Taxonomy Development

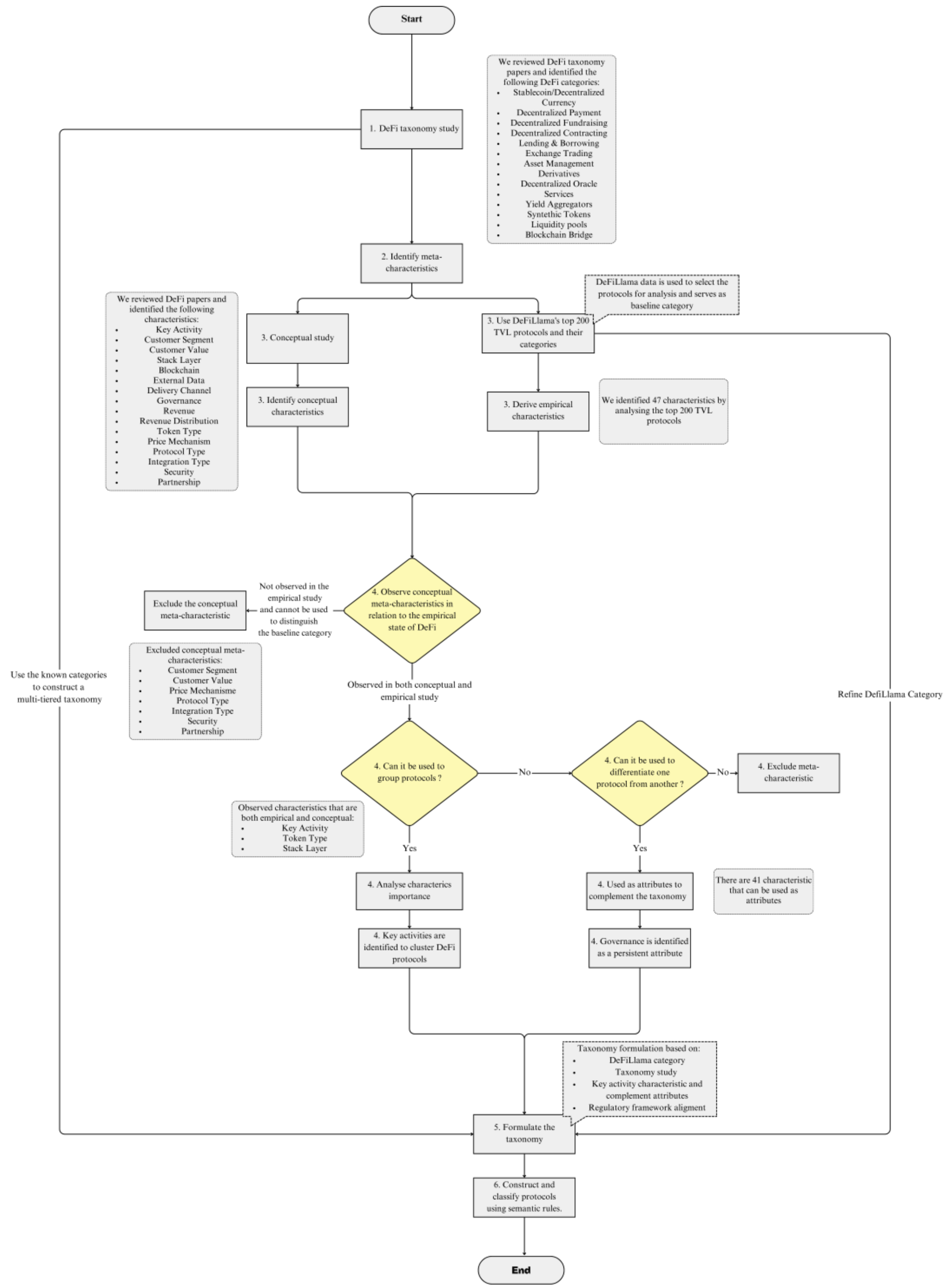

Figure 3: Methodology flowchart on formulating the taxonomy

This section details the development of our dynamic DeFi taxonomy, with the methodology illustrated in Figure 3. First, we conduct an in-depth review of existing literature on DeFi classifications and taxonomies to identify previously established categories. Second, we focus on identifying meta-characteristics to determine which traits could effectively differentiate the various DeFi categories. Third, we examine the empirical conditions of the DeFi ecosystem, analysing their web presence, implementation documentation, and governance measures. We use publicly available data from DeFiLlama as a reference, helping prioritise the protocols for further exploration.focusing predominantly on the 200 largest protocols by Total Value Locked (TVL) on Ethereum. Fourth, key meta-characteristics are then mapped to support differentiation among protocols, culminating in the construction of the taxonomy and the development of semantic rules to categorise each protocol accurately.

Taxonomy Study

Through an analysis of existing literature, as summarised in Table 2, we identify 14 overarching conceptual categories within the DeFi ecosystem, detailed in Table 2. These categories represent the primary functions and services provided by DeFi protocols, forming a foundational framework for understanding the ecosystem's diversity. Several categories, such as Lending and Borrowing, Stablecoins, Automated Asset Management, Trading/Exchange, and Payment Services, appear consistently across multiple studies (Chen and Bellavitis, 2020) (Brühl, 2021) (Moncada et al., 2021) (Schär, 2021) (Nelaturu et al., 2022) (Xu and Xu, 2022) (Anoop and Goldston, 2022) (Puschmann and Huang-Sui, 2024) (Gogol et al., 2024), highlighting their foundational importance in DeFi operations. As shown in Table 2, these categories serve as core archetypes in the DeFi ecosystem and provide a critical basis for further development of the taxonomy.

Table 2: Identified DeFi Categories.

Authors	**Identified Categories**
(Chen and Bellavitis, 2020)	Decentralised currencies, Payment, Decentralised fundraising, Decentralised contracting
(Brühl, 2021)	Payments, Savings/Loans, Trading Platforms, Capital Markets, DLT Infrastructure, Digital Financial
(Jensen et al., 2021)	Decentralised Exchanges and Automated Market Makers, Peer-to-Peer Lending and Algorithmic Money Markets, Derivatives, Automated Asset Management
(Moncada et al., 2021)	Lending & Borrowing, Exchange Trading, Asset Management, Derivatives, and Stablecoin
(Schär, 2021)	Asset Tokenization, Decentralised Exchange Protocols, Decentralised Lending Platforms, Derivatives, Asset Management
(Nelaturu et al., 2022)	Digital Identity, Payments, Digital Currencies, Decentralised Exchanges, Protocols for Loanable Funds, Derivatives, Decentralised Oracles, Decentralised Storage, Services, Corporate Governance, Crowdfunding
(Xu and Xu, 2022)	Protocols for Loanable Funds, Decentralised Exchanges, Yield Aggregators
(Anoop and Goldston, 2022)	Aggregator, Lending & Exchanges, Asset Management, Derivatives, & Others
(Puschmann and Huang-Sui, 2024)	Payment, Financing, Investment, Insurance, Trading

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Table 2: Identified DeFi Categories. (Continued)

(Beinke et al., 2024)	One-Stop-Shops, Lending & Borrowing, Yield Optimization, Ecosystem Driver, Decentralised Exchange
(Gogol et al., 2024)	Trading, Lending, Asset Management, Synthetic (Pegged) Tokens, Liquidity Pools, Aggregator, Blockchain Interoperability, Liquidity Pools

Identify Meta-Characteristics

To establish a robust classification framework, we review a comprehensive body of literature on DeFi taxonomies, prioritising meta-characteristics that not only appear consistently across multiple studies but also demonstrate clear applicability for distinguishing between DeFi protocol types. This process focuses on selecting high-level, abstract categories that could effectively capture the functional, governance, and structural nuances across diverse DeFi protocols. Through this analysis, we identify 16 meta-characteristics that have been used in previous studies, as shown in Table 4. Concurrently, we analysed empirical aspects of the DeFi ecosystem, including protocols' web presence, documentation, and governance structures, to understand how they categorise themselves. This provides a foundational baseline for our taxonomy. (DefiLlama, 2020b).

To asses the practicallity of our taxonomy, we use a dataset from DeFiLlama(DefiLlama, 2020b) comprise of more than 4,000 public DeFi protocols across 200+ different blockchains. Ethereum is chosen blockchain as sample blockchain, since many of protocols run on Ethereum blockchain also run on other blockchain such as, AAVE also run on Arbitrum, Base, Avalanche, and other blockchains. Then we prioritise Ethereum's top 200 protocols, as these represent approximately 97.96% of Ethereum's TVL. Additionally, DefiLlama also has their own categorization for each of the protocols, such as Lending, Collateralised Debt Position (CDP), Decentralised Exhcanges (DEXs) (DefiLlama, 2020a), which will be used later to refine the categorization and taxonomy.

From our empirical analysis, we identify 47 empirical characteristics, which are outlined in Table 3. We then combined these empirical characteristics with the conceptual meta-characteristics that we had identified in the top 200 protocols' web page, documentation/white paper, and 3rd party coverage of a particular protocol. Only the conceptual meta-characteristics that are observable in the top 200 DeFi protocols are included in the study, ensuring that the taxonomy remains relevant and reflective of the current DeFi ecosystem. In contrast, all derived empirical meta-characteristics is retained to capture the full empirical spectrum of the observed protocols .

Table 3: Identified Empirical Meta-Characteristics

Meta-characteristic	**Definition**
Key Activity	The business activity of a protocol.
Governance	The governance measure of a protocol.
Token Type	Type of token: native blockchain token, NFT, liquidity provision token, or restaking token.
Type of futures	Type of future instrument that is sold.
Collateral asset	Type of digital asset used as the loan collateral.
Type of interest	Type of interest rate the loan set: fixed rate, variable interest rate, or zero rate / discount rate.
Source of fund	Loan funding origin: direct p2p, pooled fund, or minted by some sort of algorithm.

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Table 3: Identified Empirical Meta-Characteristics (Continued)

Liquidation factor	Mechanism for liquidating bad loan.
Redemption/Reimbursement	How the principal amount of debt instruments is repaid by debt issuer.
Custodial	How the exchange liquidity is being held: centralised, in third party partner or non-custodial in a smart contract.
Fee structure	The structure of fee imposed by the exchange protocol: static or dynamic. Static implies constant fee structure, dynamic implies changing fee structure depending on market conditions.
Relative fee	The relative fee compared to other protocols in this category.
Blockchain	The supported blockchain of the DeFi protocol: does it operate on a single blockchain, across multiple blockchains, or route transactions between blockchains?
Price determiner	How the exchange rate is determined: automated market making (AMM), order book, or hybrid of AMM and order book.
Escrow dependency	Whether a protocol has dependencies with an escrow account.
Object	What is the target object of the service offered, i.e, smart contract, transaction, token/asset, etc.
Data feed	What kind of data that is being fed by the oracle.
Layer	The source and destination layer of the bridge asset or transaction in the DeFi stack layer, e.g. bridging native Ethereum to native Arbitrum cryptocurrency, bridge native BTC to Ethereum L1 asset as a wrapped BTC.
Processing fee currency	What currency or digital asset token used to pay the processing cost.
Privacy method	Method to achieve the desired privacy.
Leveraged asset	It indicates the type of asset, token, or liquidity mechanism being leveraged by the yield farm.
Yield Generation	It indicates the method of yield generation.
Asset class	Type of asset class accepted by the index fund.
Weighting type	Method of weighting the bundled asset class into an index.
Index return type	The offered return of the index fund.
Asset customisability	Whether the bundled assets are fixed or customisable.
Control mechanism	Method of controlling the market making activities.

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Table 3: Identified Empirical Meta-Characteristics (Continued)

Oracle dependency	This specifies whether or not the protocols required an oracle to feed external data.
Instrument dependency	This identifies the type of financial instrument that has been tokenised.
Token standard	This specifies the used token standard.
Cost structure	It shows how the cost of yield generation is passed on to the user.
Entitlement	This specifies whether the entitlement of the tokenised asset is revokable or not.
EigenLayer dependency	This specifies whether the protocols use EigenLayer solutions to restake.
Restaking type	Type of EigenLayer staking mechainsm used to restake the staked token.
Value reference	A value of an asset or currency that a stablecoin aims to maintain.
Stabilisation technique	The mechanism used to maintain a stable value.
Collateral custody	Where the collateral is stored.
Minting mechanism	The process of minting a new token.
Underlying asset	Asset or entity that a financial instrument, like a derivative, is based upon.
Type of output	The output format of a given protocol: either a tool, platform or an asset.
Option type	Type of option, either a call or a put option.
Instrument Standard	Used financial instrument standard that is aligned with CFI ISO 10962.
Delivery channel	How the service is delivered to the user.
Service offered	What kind of service a platform gives: funding/liquidity, deployment tool, or multiple forms of services.
Processing mechanism	How the transaction is processed. It involves the technical aspect of rolling up transactions, network bridging, or other novel mechanisms.

Table 4: Identified Conceptual Meta-Characteristics

Meta-characteristic	**Observed in Paper**	**Observed in DeFi Protocols**

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Table 4: Identified Conceptual Meta-Characteristics (Continued)

Key Activity	(Chen and Bellavitis, 2020),(Schär, 2021), (Moncada et al., 2021),(Eikmanns et al., 2023), (Jensen et al., 2021), (Katona, 2021), (Qin et al., 2021), (Werner et al., 2022), (Brühl, 2021), (Nelaturu et al., 2022), (Anoop and Goldston, 2022), (Xu and Xu, 2022), (Meyer et al., 2022), (Puschmann and Huang-Sui, 2024), Beinke et al. (Beinke et al., 2024), (Gogol et al., 2024)	✓
Customer Segment	(Moncada et al., 2021), (Katona, 2021), (Beinke et al., 2024)	✗
Customer Value	(Moncada et al., 2021), (Katona, 2021), (Beinke et al., 2024)	✗
Customer Value	(Moncada et al., 2021), (Eikmanns et al., 2023), (Jensen et al., 2021), (Werner et al., 2022), (Puschmann and Huang-Sui, 2024), (Beinke et al., 2024), (Gogol et al., 2024)	✗
Stack Layer	(Schär, 2021), (Eikmanns et al., 2023), (Katona, 2021), (Anoop and Goldston, 2022), Beinke et al. (Beinke et al., 2024),(Gogol et al., 2024)	✓
Blockchain	(Schär, 2021), (Moncada et al., 2021), (Katona, 2021), (Brühl, 2021), (Puschmann and Huang-Sui, 2024), (Beinke et al., 2024), (Gogol et al., 2024)	✓
External Data	(Katona, 2021), (Qin et al., 2021), (Nelaturu et al., 2022), (Meyer et al., 2022)	✓
Delivery Channel	(Eikmanns et al., 2023), (Jensen et al., 2021), (Katona, 2021), (Brühl, 2021), (Anoop and Goldston, 2022), (Beinke et al., 2024)	✓
Governance	(Eikmanns et al., 2023), (Jensen et al., 2021), (Katona, 2021), (Werner et al., 2022), (Nelaturu et al., 2022)	✓
Revenue	(Jensen et al., 2021), (Katona, 2021), (Werner et al., 2022), (Anoop and Goldston, 2022), (Xu and Xu, 2022), (Beinke et al., 2024)	✓
Revenue Distribution	(Beinke et al., 2024)	✓
Token Type	(Puschmann and Huang-Sui, 2024), (Gogol et al., 2024)	✓
Price Mechanism	(Puschmann and Huang-Sui, 2024)	✗
Protocol Type	(Puschmann and Huang-Sui, 2024), (Gogol et al., 2024)	✗
Integration Type	(Puschmann and Huang-Sui, 2024), (Gogol et al., 2024)	✗
Security	(Beinke et al., 2024)	✗

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Table 4: Identified Conceptual Meta-Characteristics (Continued)

Partnership	(Beinke et al., 2024)	✗

To ensure practical applicability we exclude categories deemed non-essential to the DeFi framework, such as Ponzi schemes, gaming, and Telegram bots, as these either lack inherent financial utility or operate outside the scope of structured DeFi services targeted for regulatory assessment. Each remaining category is then evaluated to confirm it represents a meaningful function within DeFi and is grounded in sound reasoning. For instance, 'Lending and Borrowing' is retained as a primary category due to its foundational role in DeFi, while categories like 'NFT collateralize loans' were too narrowly defined and could be grouped under more general categories. Stablecoins are included as a category to acknowledge their significant impact on DeFi, despite their exclusion in existing literature.

Simultaneously, we revisited previous DeFi research to examine category definitions. We identified six core categories that have frequently appeared in earlier studies: Lending and Borrowing, Exchange, Services, Stablecoin, Asset Management, and Liquid Staking. Some newer DeFi categories, such as insurance, options, and futures, have not been extensively studied in DeFi taxonomy research, but are well-understood within TradFi.

To address these gaps, we explored how TradFi categorises these financial services. The earliest foundational work in TradFi taxonomy can be traced to Hakansson's 1982 paper, 'To Innovate or Regulate?' (Hakansson, 1982), which emerged during a period of rapid adoption of financial instruments like options, futures, and derivatives. Hakansson's work provided the foundation for future TradFi taxonomies, leading to the International Organization for Standardisation (ISO) publishing ISO 10962 in 1997(ISO, 2021). Commonly known as the Classification of Financial Instruments (CFI), ISO 10962 offers a formal taxonomy standard that describes financial instruments using a six-letter code, categorizing instruments by type, group, and attributes. Widely adopted by international financial institutions, the CFI standard is periodically updated to reflect evolving market conditions.

Taxonomy Formulation

Building on the previously identified meta-characteristics, we develop a structured DeFi taxonomy by refining and categorising these characteristics based on their relevance to both DeFi and regulatory stakeholders. Figure 4 presents a visual overview of the taxonomy, with each primary category representing a distinct function within the DeFi ecosystem.

Drawing inspiration from TradFi's approach, we structure our DeFi taxonomy with a framework similar to ISO 10962. In our system, each DeFi protocol nature which include the activity, entity, asset, and infrastructure is classified into multi-tiered categories, with additional attributes that refine its categorization. However, rather than directly replicating ISO 10962, we incorporated unique meta-characteristics specific to DeFi. These are divided into static and dynamic attributes, each serving a distinct purpose. Dynamic attributes capture specific features relevant to certain categories or subcategories; for example, the 'Peer-to-Peer (P2P) Lending' subcategory includes attributes like collateral type, interest rate, source of funds, and liquidation criteria to assess risk exposure. In contrast, the 'Decentralised Exchanges (DEXs)' subcategory includes attributes such as custodial structure, fee model, supported blockchains, and price determination mechanisms, providing essential details for nuanced classification.

Finally, we construct the taxonomy with several guiding principles. First, we integrate both conceptual and empirical dimensions of DeFi to create a robust foundation. Second, we examine the regulatory landscape, focusing on the question, 'Who or what should be regulated?' This allows us to align each protocol with appropriate regulatory perspectives, given that the concept of a 'protocol' does not currently exist within traditional financial regulations. Additionally, we use the ISO 10962 framework as a bridge to connect DeFi and TradFi, fostering a more cohesive understanding of their potential interactions. Lastly, we aim for comprehensive coverage, capturing the diverse activities and innovations within the DeFi ecosystem to provide a holistic and inclusive view.

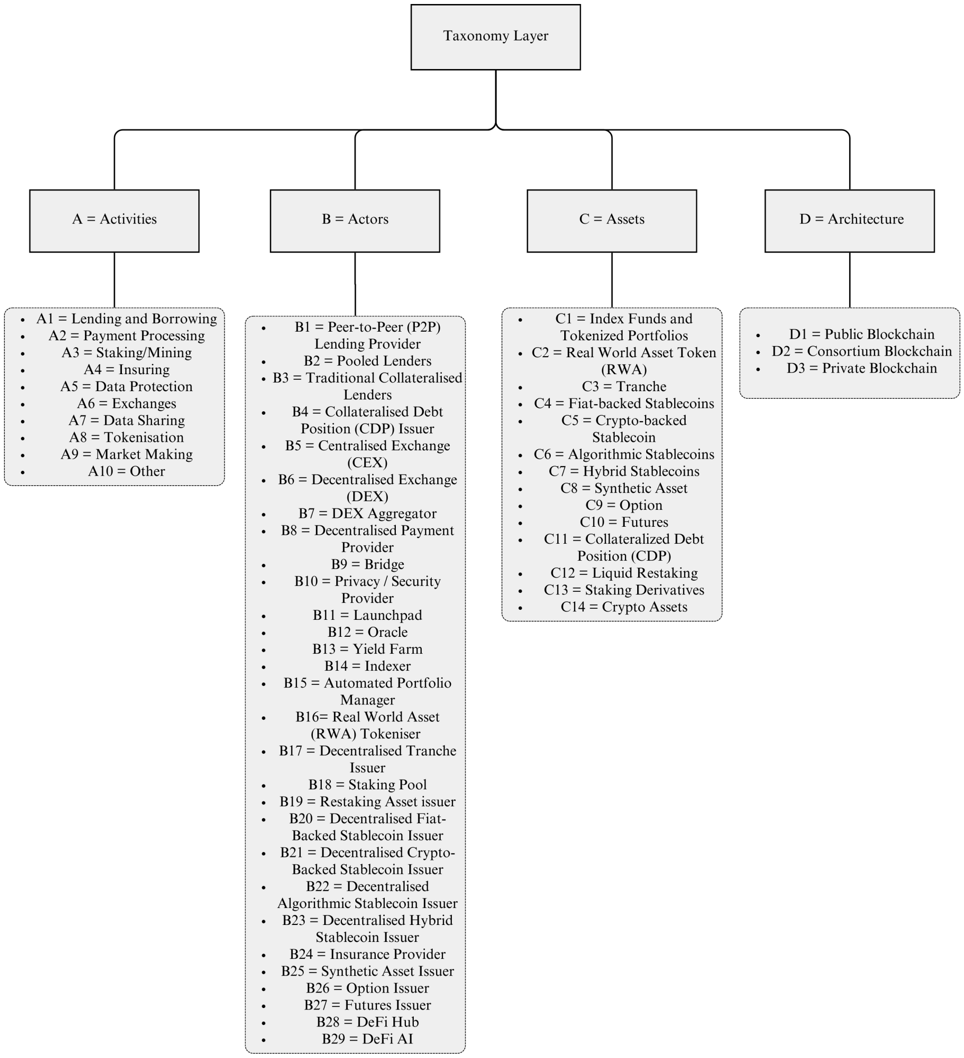

Figure 4: High Level Categorization of Our Taxonomy

Taxonomy Type

Our taxonomy begins with identifying the high-level categorization, consolidating DeFi into four primary types, referred to as the '4As': Activities, Actors, Assets, and Architecture, as depicted in Figure 4. This framework provides an abstracted view of DeFi categories that is particularly relevant for regulatory analysis, where primary risks are closely tied to the classification of these types.

By categorising DeFi protocols this manner, the taxonomy is designed to simplify oversight by clarifying key operational and financial distinctions, enabling regulators to focus on targeted compliance measures, risk assessments, and governance standards tailored to each category's unique characteristics.We conceptualise the first three categories—Activities, Actors, and Assets—as perspectives for understanding regulatory implications within DeFi. The Architecture type defining the variations in blockchain infrastructure available to date. Each category is defined as follows:

- Activities: Actions or processes performed within the DeFi ecosystem, similar to activities in TradFi.
- Actors: Intermediaries, platforms, protocols, or organisations that provide services in the DeFi ecosystem,

similar to entities in TradFi.

- Asset: Financial instruments or products that have value and can be traded, similar to assets in TradFi.
- Architecture: The underlying blockchain technology responsible for processing DeFi transactions.

Taxonomy Categories

The second layer of our taxonomy involves categorising individual DeFi protocols and alongside variations in blockchain infrastructure. This categorisation is based on an analysis of the empirical DeFi landscape, relevant literature, and publicly accessible databases. To address architectural considerations, we draw on existing blockchain research to identify key factors, such as use cases, consensus mechanisms, and permission structures (Antal et al., 2021) (Soltani et al., 2022) (Gao et al., 2018) (Belotti et al., 2019) (Monrat et al., 2019) (Zheng et al., 2018). Given our aim to create an accessible and understandable taxonomy for regulatory audiences, we categorise DeFi infrastructure based on permission mechanisms, distinguishing between public, consortium, and private blockchains. These categories are defined in Table 5, Table 6, Table 7, and Table 8 for the activity, actor, asset, and architecture type respectively.

Table 5: Activities Type Category Definition

Category	**Definition**
A1 = Lending and Borrowing	The process of issuing a loan.
A2 = Payment Processing	The process of handling transactions on a blockchain.
A3 = Staking/Mining	The process of staking or mining on a blockchain.
A4 = Insuring	The process of insuring transactions, assets, tokens on a blockchain.
A5 = Data Protection	The process of increasing the data security and privacy.
A6 = Exchanges	The process of exchanging assets or tokens on a blockchain.
A7 = Data Sharing	The process of sharing data/information from one blockchain to another blockchain or from offchain to on chain.
A8 = Tokenisation	The process of tokenising of an asset or a basket of assets.
A9 = Market Making	The process of setting the prices of an asset at a given price.
A10 = Other	Any process that does not fall into the aforementioned categories.

Table 6: Actors Type Category Definition

Category	**Definition**
B1 = Peer-to-Peer (P2P) Lending Provider	A protocol that facilitates direct lending and borrowing between individuals.

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Table 6: Actors Type Category Definition (Continued)

B2 = Pooled Lenders	A protocol that pools individual assets to generate digital assets to lend.
B3 = Traditional Collateralised Lenders	A protocol that underwrites a loan based on collateral assets.
B4 = Collateralised Debt Position (CDP) Issuer	A protocol that issues a CDP.
B5 = Centralised Exchange (CEX)	An intermediary service for the trading or exchanging of crypto assets, similar to a stock exchange.
B6 = Decentralised Exchange (DEX)	A protocol offering the trustless trading of crypto assets on a blockchain.
B7 = DEX Aggregator	A protocol that aggregates the transactions across DEXs in order to find the best trading paths, liquidity, and transaction across blockchains.
B8 = Decentralised Payment Provider	A protocol that supports the process of payment or transaction on a blockchain.
B9 = Bridge	A protocol that facilitates bridging inter blockchain assets, data, or information.
B10 = Privacy / Security Provider	A protocol that offers increased privacy to other protocols.
B11 = Launchpad	A protocol that serves as a platform for developing and deploying smart contracts, or DApps, for DeFi applications.
B12 = Oracle	A protocol that provides external data to smart contracts, including price feeds and real-world events.
B13 = Yield Farm	A protocol that facilitates the strategic allocation of assets across various DeFi protocols to maximise returns through interest, rewards, or token incentives.
B14 = Indexer	A protocol that tokenises bundles of assets into a new digital asset.
B15 = Automated Portfolio Manager	A protocol that executes (and rebalances) token portfolios based on predefined strategies.
B16= Real World Asset (RWA) Tokeniser	A protocol that tokenises versions of real-world financial assets are sold as tokens on the blockchain.
B17 = Decentralised Tranche Issuer	A protocol that issues a tranche.
B18 = Staking Pool	A protocol that pools PoS staker token to perform validation.
B19 = Restaking Asset issuer	A protocol that issues synthetic assets or tokens representing staked assets, allowing users to participate in DeFi while their assets are staked.
B20 = Decentralised Fiat-Backed Stablecoin Issuer	A protocol that maintains a stable value by managing a reserve portfolio of cash, money market instruments, and other assets.

Continued on next page

Table 6: Actors Type Category Definition (Continued)

B21 = Decentralised Crypto-Backed Stablecoin Issuer	A protocol whose value comes from issuing new units against eligible digital assets at specific penalty rates (over-collateralisation), reflecting individual volatility risk.
B22 = Decentralised Algorithmic Stablecoin Issuer	A protocol that maintains its token value stability by dynamically adjusting the supply through the issuance of seigniorage shares.
B23 = Decentralised Hybrid Stablecoin Issuer	A protocol that uses both an underlying asset and a certain algorithm or strategy to achieve a stable token value.
B24 = Insurance Provider	A smart contract that provides coverage for unlikely events, similar to insurance products in TradFi.
B25 = Synthetic Asset Issuer	A smart contract or set of smarts contract which generate money-like token based on complex algorithm or trading strategy.
B26 = Option Issuer	A smart contract that issues options on a blockhain.
B27 = Futures Issuer	A smart contract that issues futures on a blockhain.
B28 = DeFi Hub	A protocol that offers multiple applications, protocols, and services.
B29 = DeFi AI	A protocol that provide AI solutions run on top of a blockchain.

Table 7: Assets Type Category Definition

Category	**Definition**
C1 = Index Funds and Tokenised Portfolios	A tokenised portfolio of digital assets similar to index of stocks or commodities.
C2 = Real World Asset Token (RWA)	A tokenised version of a real-world financial asset that can be sold as a token on the blockchain.
C3 = Tranche	A security product that groups various investments, sold based on risk appetite or expected return.
C4 = Fiat-backed Stablecoins	A digital asset that maintains a stable value relative to a fiat currency, by managing a reserve portfolio of cash, money market instruments, and other assets.
C5 = Crypto-backed Stablecoin	A digital asset that maintains a stable value relative to another cryptoasset and maintains this stability by issuing new units against eligible cryptoassets, with over-collateralisation to account for individual volatility risk.

Continued on next page

Table 7: Assets Type Category Definition (Continued)

C6 = Algorithmic Stablecoins	A digital asset that maintains a stable value by dynamically adjusting supply, often through the issuance of seigniorage shares.
C7 = Hybrid Stablecoins	A digital asset that maintains value stability using both underlying assets and an algorithm or strategy for achieving a stable token value.
C8 = Synthetic Asset	An asset based on a smart contract or a set of smarts contracts which generate money-like tokens based on complex algorithms or trading strategies.
C9 = Option	An asset based on a smart contract similar to TradFi options.
C10 = Futures	An asset based on a smart contract similar to TradFi TradFi futures.
C11 = Collateralised Debt Position (CDP)	An asset generated from a specific no interest collateralised loan, with liquidation being triggered below a specific loan-to-value (LTV).
C12 = Liquid Restaking	A synthetic asset or token representing staked assets, allowing users to participate in DeFi while their asset are staked using EigenLayer protocols.
C13 = Staking Derivatives	A synthetic asset or token representing staked assets, allowing users to participate in DeFi while their asset are staked, with restaking methods that do not use EigenLayer protocols.
C14 = Crypto Asset	A decentralised medium of exchange which use cryptographic measure to ensure security in the creation process and prevent double spending.

Table 8: Architecture Type Category Definition

Category	**Definition**
D1 = Public blockchain	Blockchain technology that allows open participation in various roles, such as blockchain validators, smart contract developers, and users.
D2 = Consortium Blockchain	Blockchains technology that allows limited public participation, whilst the administrator/validator roles are reserved for the group of the blockchain's promoters.
D3 = Private Blockchain	Blockchain technology that is exclusively available to approved entities.

Static Attributes

DeFi governance measure is an important aspect of DeFi as all DeFi protocols have some sort of mechanism of governance and it provides meaningful insight on DeFi state of governance risk (Aramonte et al., 2021). Unlike TradFi, where governance is typically carried out through board meetings, DeFi governance adopts a moreS decentralised model. In the DeFi ecosystem, all protocols have some sort of governance method: on chain through governance token issuance, off chain method in discussion forum or hybrid where initial proposal it discussed first in forum then the voting is held on chain to reach a community consensus.In on-chain governance, any holder of governance tokens has a degree of voting rights and can participate in decision-making. Since voting power is based on the ownership of governance tokens, the distribution of these tokens significantly impacting the governance outcomes and roadmap Although it is expected for any DeFi protocol to be fully decentralised, we found over 90% the DeFi protocols that have 0.99 Gini Index (Sen, 1997), which indicate the presence of risk associated with DeFi centralisation (Aramonte et al., 2021).

We organise this information DeFi governance concentration as granular categorisation to provide better view of the risks related to centralisation within a DeFi protocol, as the potential systemic risk associated with centralisation in DeFi governance is significant (Aramonte et al., 2021). We breakdown the centralisation of DeFi governance using the wealth inequality Gini coefficient (Sen, 1997), alongside the governance method, as shown in Table 9.

Table 9: Static Attribute: Governance Concentration & Governance Method Variations

Governance Concentration	**Governance Method**
F = Fully centralised	O = On-chain governance
I = Illusionary Decentralised (Governance token distribution gini ¿0,8)	F = Off-chain Governance
S = Semi Decentralised (Governance token distribution gini 0,6-0,8)	H = Hybrid Governance
D = Decentralised (Governance token distribution gini ¡0,6)	X = Not Appl./Undefined
X = Not Appl./Undefined	

Dynamic Attributes

Dynamic attributes are chosen based on empirical meta-characteristics derived from analysing the top 200 protocols by Total Value Locked (TVL). These attributes capture protocol-specific functionalities and operational nuances, such as collateral structure, governance mechanisms, and asset interoperability, which are essential for providing a granular view of the DeFi ecosystem. They enable the taxonomy to reflect real-world usage patterns and adapt to protocol-level differences that static categories may not fully address. We have defined each dynamic attribute as in Table 3.

Proposed Taxonomy Structure

When designing our taxonomy structure, as shown in Figure 5, we also considered how this information could be transmitted seamlessly and be compatible with traditional financial (TradFi) infrastructure. Drawing inspiration from CFI ISO 10962, we recognised that it can be transmitted through the Financial Information eXchange (FIX) protocol, the de facto messaging standard for pre-trade and trade communication in global equity markets (FIX, 2004). The FIX protocol is a machine-readable format for data exchange, where each piece of information has its own tag and value, including CFI ISO 10962. In the case of CFI ISO 10962, each piece of information is codified to allow for fast data exchange. Therefore, following this structure to codify the taxonomy enables easy integration and data exchange within TradFi infrastructure.

The structure of our taxonomy as a stack layer categorisation starting from the top level specifying the DeFi activity. We then abstract the DeFi protocol as the actor with its own static and dynamic attributes, this abstraction of DeFi protocol is specified under the appendix in Figure 7, Figure 8 and Figure 9. Next we identify which asset involved and the type of block chain architecture it built upon. The general structure of our taxonomy is outline in Figure 5.

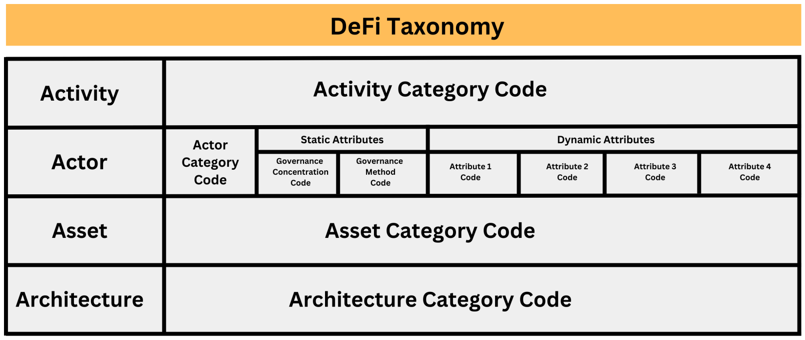

Figure 5: General Taxonomy Structure

To illustrate our taxonomy structure, consider AAVE—a leading DeFi protocol providing lending and borrowing services—which will have four sets of tags representing activity, actor, asset, and architecture. First, the taxonomy will assign the activity category lending and borrowing code "A1" to the activity tag. Next, the actor tag value is represented by a codified 8-9 character string. In the case of AAVE, the value is "B2IOMVRA", where the string "B2" signifies the pool-lending actor category, the next two characters represent governance concentration and governance method, respectively, and the final four characters capture the dynamic attributes: collateral asset, interest type, source of funds, and liquidation mechanism. The asset tag and architecture tag values are code "C14" for crypto currency asset category and "D1" for public block chain, respectively.

Semantic Rules and Classification

To operationalize the taxonomy, we developed semantic rules for protocol classification. These rules were constructed by extracting keywords from protocol documentation, including welcome pages, white papers, technical descriptions, and governance sections. A detailed explanation of this process is presented in the semantic rule flowchart Figure 6.

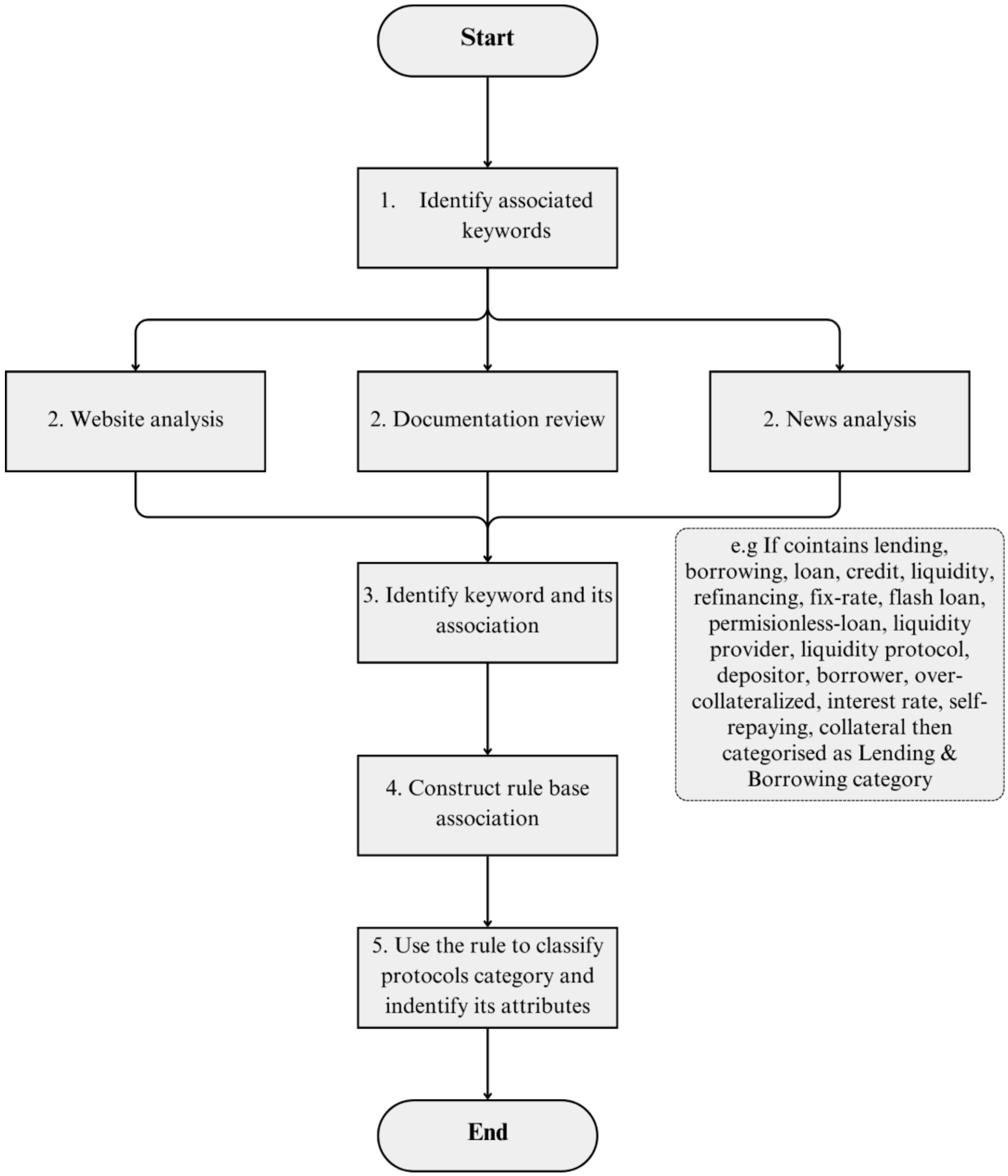

Figure 6: Semantic Rule Development Flowchart

The flowchart illustrates a systematic approach for classifying protocols through a semantic rule-based methodology. This process begins with the extraction of relevant keywords from various authoritative sources associated with each protocol. Specifically, these keywords are derived from the protocol's welcome page, white paper, documentation overview, technical functionality descriptions, and governance sections. These identified keywords are then used to construct rule-based associations that categorize the protocol into a specific category and sub-category. This approach leads to an efficient categorization of the top 200 TVL protocols and achieves consistent results.

For instance, in the case of Aave, a prominent lending and borrowing protocol, the landing page features keywords such as "Liquidity Protocol," "Liquidation Fees," "Borrow," and "Supply," which indicate its classification as a "Pool-based Lending" protocol. Further examination of Aave's documentation reveals additional terms, including "Overcollateralised," "Interest rate model," and "Non-custodial liquidity protocol," further substantiating its categorization. Additionally, external sources such as news articles contribute supplementary keywords, such as "Unified Liquidity Layer," "Liquidation Engine," and "Chainlink Oracles," which collectively reinforce the identification of Aave within the Pool-based Lending category.

Guidance and Context

This study presents a taxonomy framework grounded in both the conceptual foundations and empirical realities of the DeFi ecosystem. The primary objective is to develop a uniform and comprehensive taxonomy that aligns with the infrastructure of TradFi. However, given the dynamic nature of DeFi markets and its community perception on some of DeFi protocols and measurement, we would like to highlight some of these matters, such as TVL double counting, interesting categories, and our contribution, with intention to offer readers a clear understanding of the taxonomy's structure and the context within which it was developed.

DeFi TVL

Within the DeFi community, there is growing concern of TVL as an ideal metric for assessing DeFi valuation. This concern arises because TVL calculates the inflow of digital assets into DeFi protocols without differentiating the types of assets involved. Often, assets counted in TVL are derived from other assets whose value has already been included, leading to potential double-counting.

For example, stETH is a token that represents staked ETH in Lido. When a user stakes ETH in Lido, they receive the equivalent value in stETH (e.g., staking 10 ETH yields 10 ETH worth of stETH), which is then included in Lido's TVL. Since stETH is a liquid asset, it can also be used as collateral in other protocols, such as AAVE. If the same user stakes ETH in Lido and subsequently uses stETH as collateral on AAVE, the stETH will be counted both in Lido's and in AAVE's TVL. This is commonly considered as double-counting, as only 10 ETH originally entered the DeFi ecosystem, but the current TVL methods record it multiple times: every time that is staked and then again when is used as collateral, and so on.

Many observers argue that this double-counting misrepresents DeFi's true valuation. Although assessing TVL's suitability as a metric is beyond the scope of this study, we use it here as a practical measure for protocol prioritisation and influence ranking within the DeFi ecosystem. Future research could explore alternative metrics, such as adjusted TVL, which accounts for collateral reuse, offering a more accurate view of DeFi's economic activity.

Noteworthy Type and Categories

Despite most previous DeFi taxonomies acknowledging blockchain as the foundational infrastructure, they often neglect to address the specific operational details of individual blockchains, assuming a universal understanding of these technical nuances. This assumption has, in many cases, exacerbated the gap between DeFi and traditional finance (TradFi). Given that blockchain technology itself is relatively new, this lack of clarity further hinders broader understanding and adoption. In contrast, our proposed taxonomy is more inclusive, addressing the critical aspects of DeFi infrastructure to bridge this gap and facilitate the integration of DeFi with TradFi.

This classification simplifies the regulatory approach for each category. Public blockchains allow open participation in various roles, such as blockchain validators, smart contract developers, and users. Regulatory bodies can, therefore, focus on regulating the types of activities permitted on these platforms. In contrast, consortium and private blockchains restrict access to specific roles, with consortium blockchains allowing limited public participation, while private blockchains are exclusively available to approved entities. For these types, regulators can focus on applying frameworks that govern the participating entities or actors, ensuring a tailored approach to regulation across different blockchain infrastructures.

Although stablecoins are often excluded from DeFi-specific taxonomies due to their centralised elements, we recognise here their critical role within the DeFi ecosystem, where they underpin liquidity, lending, and payment services. Stablecoins have largely contributed and continue to facilitate stability and interoperability across DeFi protocols, providing an essential bridge between TradFi and DeFi assets. For this reason, stablecoins are divided into four subcategories within our taxonomy to better reflect their range of diverse applications and regulatory implications.

While centralised exchanges (CEXs) operate outside of DeFi due to their custodial nature and centralised management, they are included in this taxonomy to acknowledge their role as critical entry point for users accessing DeFi platforms. Additionally, CEX data often serves as a source for oracles within DeFi applications, underscoring an indirect yet pivotal influence on DeFi operations.

Usage Guidance

This taxonomy research is part of the Cambridge Centre for Alternative Finance (CCAF) digital asset research. This taxonomy represents a foundational step in mapping the DeFi ecosystem, with a particular emphasis on accessibility for regulators, industry practitioners, and academic researchers. By making this data available through the CCAF's Cambridge DeFi Navigator, we aim to foster a collaborative approach to DeFi research and regulatory understanding. Open access allows a broad range of stakeholders to contribute insights, validate categorisations, and support real-time analysis, creating a dynamic resource that evolves alongside DeFi innovations. We aim to provide a clear understanding of the entire DeFi ecosystem, as of now we have mapped top 200 TVL protocols in Ethereum covering 97.96% of the entire Ethereum TVL. Subsequently, we will expand the coverage through not just limiting it to the Ethereum blockchain but many more available blockchains.

Contribution

Despite extensive research on DeFi taxonomies, current studies lack a standardised approach and fail to establish a direct connection between DeFi and TradFi. There is a clear gap in the literature for a DeFi taxonomy that aligns with existing TradFi infrastructure while remaining accessible to regulators and practitioners. Most academic research has treated TradFi and DeFi as separate domains, examining their structures, risks, and innovations independently. Although much has been studied about DeFi protocols and TradFi regulatory frameworks, the critical intersection where traditional financial practices meet decentralised technologies remains largely unexplored. This gap has created a significant void in understanding how to effectively integrate TradFi and DeFi, which is crucial for harnessing the strengths of both systems while addressing the unique challenges their convergence presents.

This study aims to address these gaps by examining DeFi protocols through the lens of financial instruments, aligning with regulatory perspectives familiar in TradFi. We propose that the current state of DeFi reflects the evolution of financial instruments in the 1970s and 1980s, a period marked by similar regulatory and acceptance challenges, inspired and learned by the regulations emerged of that era. These historical parallel underscores the need for a clear, practical taxonomy that enables stakeholders to understand DeFi protocols within a regulatory context. Our taxonomy is designed to be both practical and compatible with existing TradFi infrastructure, facilitating integration without significant backend development costs for connecting DeFi with TradFi. This approach creates a cohesive framework that bridges the gap between decentralised and traditional financial systems.

Future Works

The taxonomy proposed in this paper provides a foundation for several future developments. One key direction is the use of this framework for AI-based automatic classification of DeFi protocols. By integrating machine learning algorithms with the dynamic attributes defined in the taxonomy, AI systems can automatically categorize new and existing DeFi protocols, significantly enhancing the scalability and accuracy of DeFi ecosystem analysis. Furthermore, the taxonomy can be leveraged to assess the risks associated with DeFi protocols by identifying critical attributes, such as governance structure, liquidity mechanisms, and asset management strategies. These attributes can serve as indicators to flag potential vulnerabilities or compliance issues, aiding in the development of more robust risk management systems. Lastly, this framework will support regulators and industry practitioners by providing a standardised benchmarking tool for evaluating and understanding DeFi protocols, enabling more informed regulatory decisions and fostering better collaboration between decentralised and traditional financial systems.

Conclusion

We propose a comprehensive multi-tier taxonomy with static and dynamic attributes for DeFi, aiming to bridge the gap between DeFi and TradFi. The proposed taxonomy is intentionally designed to harmonise DeFi classifications with TradFi standards, particularly ISO 10962, by adopting a multi-layered framework that addresses both functional and structural distinctions within DeFi protocols. Leveraging ISO 10962 as a foundation it facilitates a smoother resource efficient regulatory integration, and enhances the taxonomy's accessibility for stakeholders familiar with this traditional financial classification, thereby fostering broader understanding and acceptance of DeFi within established financial infrastructures.

Our taxonomy is based on key activities and governance structures combined with dynamic attributes representing the detail of DeFi complexity, which are only relevant for specific subcategory. Our taxonomy covers a broad spectrum of DeFi applications such as lending, exchanges, and liquid staking. Additionally, we recognize the role of stablecoins, centralised exchanges, and other subcategories, which have largely contributed to the DeFi adoption and further reflect the evolving complexity of DeFi. Finally, we showed classification of the top 200 TVL DeFi protocol using our taxonomy framework.

Appendix

DeFi Category		B1 = Peer-to-Peer (P2P) Lending Provider	B2 = Pooled Lenders	B3 = Traditional Collateralised Lenders	B4 = Collateralised Debt Position (CDP) Issuer	B5 = Centralised Exchange (CEX)	B6 = Decentralised Exchange (DEX)	B7 = DEX Aggregator	B8 = Decentralised Payment Provider	B9 = Bridge	B10 = Privacy / Security Provider
Static Attribute	Governance Concentration	Governance Concentration: F = Fully Centralised, I = Illusionary Decentralised (Governance token distribution gini >80), S = Semi Decentralised (Governance token distribution gini 60-80), D = Decentralised (Governance token distribution gini <60) X = Not Appl./Undefined									
	Governance Method	Governance Method: O = On-chain Governance, F = Off-chain Governance, H = Hybrid Governance X = Not Appl./Undefined									
Dynamic Attribute	Attribute 1	Collateral Asset: S = Stablecoin R = Real World Asset D = Non stablecoin digital asset H = Hybrid	Collateral Asset: S = Stablecoin R = Real World Asset D = Non stablecoin digital asset H = Hybrid	Collateral Asset: S = Stablecoin R = Real World Asset D = Non stablecoin digital asset H = Hybrid	Collateral Asset: S = Stablecoin R = Real World Asset D = Non stablecoin digital asset H = Hybrid	Custodial: C = Centralised T = Third's Party N = Non-Custodial	Custodial: C = Centralised T = Third's Party N = Non-Custodial	X = Not Appl./Undefined	Chain: S = Single chain O = Omnichain X = Not Appl./Undefined	Use Escrow: Y = Yes N = No X = Not Appl./Undefined	Object: S = Smart contract T = Transaction X = Not Appl./Undefined
	Attribute 2	Type of interest F = Fixed rate Z = Zero rate / discounted rate V = Variable interest rate	Type of interest F = Fixed rate Z = Zero rate / discounted rate V = Variable interest rate	Type of interest F = Fixed rate Z = Zero rate / discounted rate V = Variable interest rate	X = Not Appl./Undefined	Fee structure: D = Dynamic Fee S = Static Fee	Fee structure: D = Dynamic Fee S = Static Fee	X = Not Appl./Undefined	Layer: N = Native - Native L1 = Native - L1 L2 = L1 - L2 M = Multilayer X = Not Appl./Undefined	Layer: N = Native - Native L1 = Native - L1 L2 = L1 - L2 M = Multilayer X = Not Appl./Undefined	Delivery Channel: W = Wallet A = API Only S = SDK Only M = Multiple X = Not Appl./Undefined
	Attribute 3	Source of Fund: P = Peer 2 Peer A = Algorithmic R = Reserve/Pool	Source of Fund: P = Peer 2 Peer A = Algorithmic R = Reserve/Pool	Source of Fund: P = Peer 2 Peer A = Algorithmic R = Reserve/Pool	Redemption/Reimbursement F = Fixed Maturity G = Fixed Maturity with Call Feature C = Fixed Maturity with Put Feature D = Fixed Maturity with Put and Call A = Amortization Plan B = Amortization Plan with Call Feature T = Amortization Plan with Put Feature L = Amortization Plan with Put and Call P = Perpetual Q = Perpetual with Call Feature R = Perpetual with Put Feature E = ExtendibleP = Perpetual F = Fixed Maturity	Relative Fee: H = High M = Medium L = Low	Chain: S = Single chain O = Omnichain R = Chain Routing	X = Not Appl./Undefined	Mechanism: R = Rollup B = Network Bridge O = Other X = Not Appl./Undefined	Processing fee currency: N = Native blockchain token G = Generated token C = Crypto currency X = Not Appl./Undefined	Privacy Method: Z = ZK-proof O = Chain Obfuscation X = Not Appl./Undefined
	Attribute 4	Liquidation Factor: T = Threshold of LTV D = Dynamic Threshold of LTV A = Algorithmic / Formula	Liquidation Factor: T = Threshold of LTV D = Dynamic Threshold of LTV A = Algorithmic / Formula	Liquidation Factor: T = Threshold of LTV D = Dynamic Threshold of LTV A = Algorithmic / Formula	Liquidation Factor: T = Threshold of LTV D = Dynamic Threshold of LTV A = Algorithmic / Formula	X = Not Appl./Undefined	Price Determiner: A = Automated Market Maker (AMM) O = Order Book H = Hybrid	X = Not Appl./Undefined	X = Not Appl./Undefined	Delivery Channel: A = App / WebApp P = API Only S = SDK Only M = Multiple X = Not Appl./Undefined	X = Not Appl./Undefined

Figure 7: DeFi Protocol Actor Category and Attributes part-1.

DeFi Category		B11 = Launchpad	B12 = Oracle	B13 = Yield Farm	B14 = Indexer	B15 = Automated Portfolio Manager	B16= Real World Asset (RWA) Tokeniser	B17 = Decentralised Tranche Issuer	B18 = Staking Pool	B19 = Restaking Asset issuer
Static Attribute	Governance Concentration	Governance Concentration: F = Fully Centralised, I = Illusionary Decentralised (Governance token distribution gini >80), S = Semi Decentralised (Governance token distribution gini 60-80), D = Decentralised (Governance token distribution gini <60) X = Not Appl./Undefined								
	Governance Method	Governance Method: O = On-chain Governance, F = Off-chain Governance, H = Hybrid Governance X = Not Appl./Undefined								
Dynamic Attribute	Attribute 1	Object: S = Smart Contract T = Token D = DApp / Decentralised App X = Not Appl./Undefined	Data Feed: C = Crypto currencey - crypto currencey price F = Fiat currency - crypto currencey price C = Commodity price D = Cross chain data	Liquidity Source: P = Crpto asset / Fund Pool L = LP Token D = Decentralised Exchange (DEX)	Asset class: S = Stablecoin D = Non stablecoin digital asset R = Real World Asset	Token Standard: A = ERC-20 B = ERC-721 C = ERC-1155 M = Multiple token O = Other token standard X = Not Appl./Undefined	Instrument Dependency: D = Debt Instrument E = Equities C = Collective Investment Vehicles R = Entitlement (Rights) O = Listed Options F = Futures T = Referential instruments L = Financing S = Swaps H = Non-listed and complex listed options I = Spots J = Forwards K = Strategies X = Not Appl./Undefined	Asset Composition: S = Stablecoin C = Cryptocurrency R = Tokenised Real World Asset M = Commodities F = Fiat Currency H = Hybrid	Custodial: C = Centralised T = Third's Party N = Non-Custodial	Custodial: C = Centralised T = Third's Party N = Non-Custodial
	Attribute 2	Service Offered: F = Funding / Liquidity D = Deployment M = Multiple	X = Not Appl./Undefined	Yield Generation: T = Trading strategy S = Staking Derivative O = Tokenisation	Weighting type: P = Price Weighted C = Capitalisation Weighted E = Equal Weighted F = Modified Market Capitalisation Weighted M = Others (Misc.)	Oracle Dependency: Y = Yes N = No	Token Standard: A = ERC-20 B = ERC-721 C = ERC-1155 M = Multiple token O = Other token standard X = Not Appl./Undefined	X = Not Appl./Undefined	Oracle Dependency: Y = Yes N = No	Eigen Layer Dependency: Y = Yes N = No
	Attribute 3	Processing fee currency: N = Native blockchain token G = Generated token C = Crypto currency X = Not Appl./Undefined	X = Not Appl./Undefined	Cost Structure: P = Periodically based on mark to market value R = Revenue percentage A = Percentage of AUM X = Not Appl./Undefined	Index return type: P = Price Return N = Net Total Return G = Gross Total Return M = Others (Misc.)	Chain: S = Single chain O = Omnichain	Cost Structure: P = Periodically based on mark to market value R = Revenue percentage A = Percentage of AUM X = Not Appl./Undefined	X = Not Appl./Undefined	Delivery Channel: P = Platform / Wallet C = API/SDK M = Multiple	Restaking Type: N = Native Restaking L = Liquid Restaking D = Delgation X = Not Appl./Undefined
	Attribute 4	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	Asset Structure: C = Customable F = Fixed	Control Mechanism: M = Modular O = Objective Function	Entitlement: R = Revokeable N = Non-revokeable X = Not Appl./Undefined	X = Not Appl./Undefined	Chain: S = Single Chain O = Omnichain	X = Not Appl./Undefined

Figure 8: DeFi Protocol Actor Category and Attributes part-2.

DeFi Category		B20 = Decentralised Fiat-Backed Stablecoin Issuer	B21 = Decentralised Crypto-Backed Stablecoin Issuer	B22 = Decentralised Algorithmic Stablecoin Issuer	B23 = Decentralised Hybrid Stablecoin Issuer	B24 = Insurance Provider	B25 = Synthetic Asset Issuer	B26 = Option Issuer	B27 = Futures Issuer	B28 = DeFi Hub	B29 = DeFi AI
Static Attribute	Governance Concentration	Governance Concentration: F = Fully Centralised, I = Illusionary Decentralised (Governance token distribution gini >80), S = Semi Decentralised (Governance token distribution gini 60-80), D = Decentralised (Governance token distribution gini <60) X = Not Appl./Undefined									
	Governance Method	Governance Method: O = On-chain Governance, F = Off-chain Governance, H = Hybrid Governance X = Not Appl./Undefined									
Dynamic Attribute	Attribute 1	Value Reference: F = Fiat currency M = Commodity C = Cryptocurrency B = Basket				X = Not Appl./Undefined	Underlying Asset: O = Option F = Futures L - Liqudity Token / Staking Token C = Cryptocurrency I = Indices W = Swaps M = Multiple	Type: C = Call option P = Put option	Type: D = DeFi Futures T = Tradfi Futures	X = Not Appl./Undefined	X = Not Appl./Undefined
	Attribute 2	Stabilisation technique: T = Tokenisation of collateral I = Interest rates on deposits and loan C = Currency Interventions O = Open market operations D = Dynamic mining reward				X = Not Appl./Undefined	Chain: S = Single chain O = Omnichain	Exercise option style A = American E = European B = Bermudan	Underlying asset: S = Stablecoin C = Cryptocurrency R = Tokenized Real World Asset M = Commodities F = Fiat Currency H = Hybrid	X = Not Appl./Undefined	X = Not Appl./Undefined
	Attribute 3	Collateral custody: S = Smart contract B =Bank Account F = Fund Manager X = Not Appl. / Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	Type: T = Asset Creation Tool A = Asset	Underlying asset: S = Stablecoin C = Cryptocurrency R = Tokenized Real World Asset M = Commodities F = Fiat Currency H = Hybrid	Standardised / Non-Standardised S = Standardised N = Non-Standardised	X = Not Appl./Undefined	X = Not Appl./Undefined
	Attribute 4	Minting Mechnism: P = Proposal D = Deposit X = Not Appl. / Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined	Standardised / Non-Standardised S = Standardised N = Non-Standardised	X = Not Appl./Undefined	X = Not Appl./Undefined	X = Not Appl./Undefined

Figure 9: DeFi Protocol Actor Category and Attributes part-3.

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